

Why can't ocean water be used for cooling AI data centers?

Posted by u/russianindianqueen • 0 points • 17 comments

Like I understand it would have logistics problems and the water would need to be filtered, but those are problems that could be easily solved. It seems like the whole AI data centers using up drinking water thing is poor planning for the long term

Comments

u/EVE_Burner_Account • 1 points • 2026-01-24 03:55 UTC

ocean water doesnt even make sense to cool nuclear reactors, why do you think it makes sense to cool data centers which are relative blocks of ice?

u/cradleu • 1 points • 2026-01-23 21:53 UTC

Because the kind of cooling that uses large amounts of water is evaporating that water. When the water evaporates, you're leaving behind a salt brine that will clog and corrode everything

u/mambotomato • 1 points • 2026-01-23 21:44 UTC

Data centers aren't actually using that much water, on a broad economic level. It's a big talking point on the internet, but like... it's simply much more affordable to divert some water from an existing river.

If you want to save water, campaign against golf courses.

u/Leverpostei414 • 1 points • 2026-01-23 18:47 UTC

There are easier and more general ways to use a minimal amount of water

u/AzorAhai1TK • 1 points • 2026-01-21 22:16 UTC

They don't use up barely any drinking water. Your whole question is based on a false premise

u/PansophicNostradamus • 1 points • 2026-01-21 00:47 UTC

Salt Corrosion. You'd need a desalination plant to extract the salt, in order for the effort to be worthwhile. Then you could use the water, but at great expense.

u/wenevergetfar • 1 points • 2026-03-02 02:50 UTC

Oh no a great expense, just think of the billionaires wallets ❤️

u/Eldergoth • 1 points • 2026-01-21 00:33 UTC

The salt would destroy the cooling system.

Massive desalination plants would need to be built to make the water viable.

You can't use salt water in steam engine electrical power plants, this is why Saudi Arabia needs desalination plants near their power plants.

u/JustUseCommonSense10 • 1 points • 2026-01-21 00:16 UTC

Ocean water has different mineral and organic compounds that affect the electrical conductivity to such a degree that these compounds must be removed before using it for cooling or your risk catastrophic damage and short-circuiting. You also have to worry about bacterial growth as well, because ocean water isn't clean.

They are likely filtering it and deionizing it.

u/KinkyCouple2323 • 1 points • 2026-01-21 00:11 UTC

I remember reading something about microsoft working on this.

u/Delehal • 1 points • 2026-01-20 23:54 UTC

There are a few problems with seawater. As other users have commented, it tends to be corrosive, and it tends to leave behind salt deposits, which can cause damage and clogging for pumps, pipes, cooling systems, and so on. You probably don't want seawater going directly into the computers, where those problems would be especially bad, but you could avoid that by using a closed-loop cooling system (similar to what's already done with nuclear power plants).

Pump and pipe equipment for saltwater costs a **lot** more to install and maintain, as I understand it.

There are a few data centers that are currently using seawater for cooling, or have looked into doing so. It's a lot easier if they are at a coastal location, so that they can have nearby intake and output pipes. That has other considerations, though, such as cost, impact on sea life, or the impact of the sea, weather, and flooding on your data center.

Finally, a lot of data centers rely on evaporative cooling, which cools off the water by letting it evaporate into the atmosphere. That's why they **use** water instead of just letting it loop around in their system forever. This doesn't work well with seawater, because the water evaporates and leaves behind annoying byproducts such as brine that can become a toxic waste problem when used in industrial quantities.

All the problems I mentioned above have potential solutions... but it's a cost and challenge that some companies would rather avoid.

u/Skatingraccoon • 2 points • 2026-01-20 23:50 UTC

We can. Microsoft did an experiment called Project Natick that showed that ocean-based cooling systems are effective, and the failure rate of their servers was actually lower, too. But they are not currently following up on this experiment and no one really knows why. Probably because of the cost of installing it and maintaining it and connecting it to the global infrastructure (that would be my guess).

u/Expensive-Body7530 • 1 points • 2026-01-23 20:59 UTC

Lol, good guess.

u/Patient-Dot-252 • 1 points • 2026-01-20 23:41 UTC

Ocean water can be used to cool AI systems. The reason it's not universally used is that it's only practical in certain locations and it introduces engineering, reliability, environmental, and permitting

constraints that often outweigh the cost savings versus conventional cooling.

u/tmahfan117 • 2 points • 2026-01-20 23:40 UTC

They're not near the ocean often, and, you don't want anything besides treated fresh water because salt water or other debris would absolutely rust, rot, clog, or otherwise destroy the cooling systems